

Date: 13/05/2016, Friday

Time: 10:00 am to 1:00 pm

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q.1 Attempt any TWO of the following:

- A What are stereo isomers? Discuss the role of stereoisomerism in pharmacological activities of natural drugs giving suitable examples. 10
- B Discuss the role of UV-visible and IR spectroscopy in the structure elucidation of phytoconstituents. 10
- C Make a note on sources, pharmacology and uses of following recent novel molecules of natural origin: 10
 - 1. Taxol
 - 2. Silymarin

Q.2 Attempt any TWO of the following:

- A Define tissue culture. Make a note on nutritional medium in context to plant tissue culture. 10
- B **Answer the following:** 10
 - 1. Discuss the advantages and disadvantages of plant tissue culture.
 - 2. Enumerate various types of plant tissue culture and discuss hairy root culture.
- C Briefly discuss the applications of plant tissue culture. Describe protoplast fusion technique and discuss its applications. 10

SECTION- II

Q.3 Attempt any TWO of the following:

- A Discuss the role of chromatographic methods in the evaluation of herbal drugs. 10
- B (i) Compare HPTLC and HPLC methods with respect to evaluation of herbal drugs. 5
- (ii) Discuss the importance of ash values and extractive values and the general procedures for determining these values. 5
- C Enumerate the parameters for evaluation of herbal drugs as per WHO guidelines and discuss the role of organoleptic and microscopical evaluation. 10

Q.4 Attempt any TWO of the following:

- A "Standardization of herbal drugs is a challenging task." Discuss the statement giving suitable examples wherever applicable. 10
- B Classify various marine natural drugs. Make a detailed note on anti cancer drugs of marine origin with examples. 10
- C Make a detailed note on anti viral and cardiovascular drugs of marine origin with examples. 10